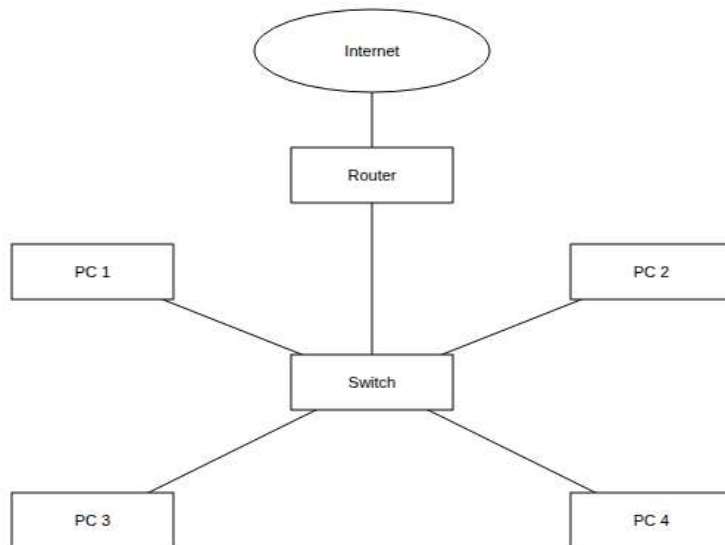


Task 1: Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

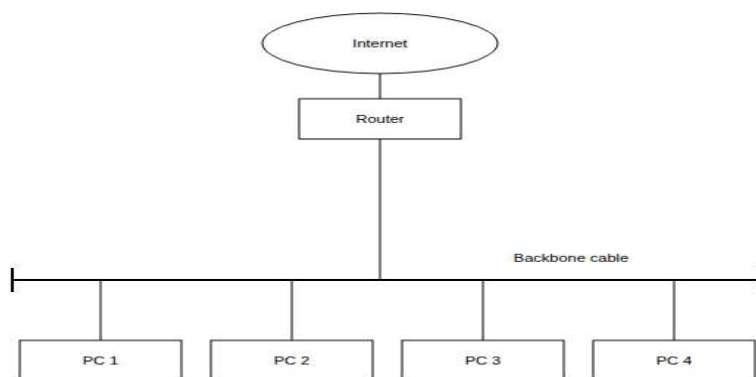
Star Topology



A central hub/switch connects individually to each device via a dedicated cable.

Real-world example: A cybercafe or small college computer lab commonly uses star topology – each PC connects individually to a central switch. If one cable or PC fails, the rest of the network keeps working, and it is easy to add or troubleshoot a single machine without disrupting others.

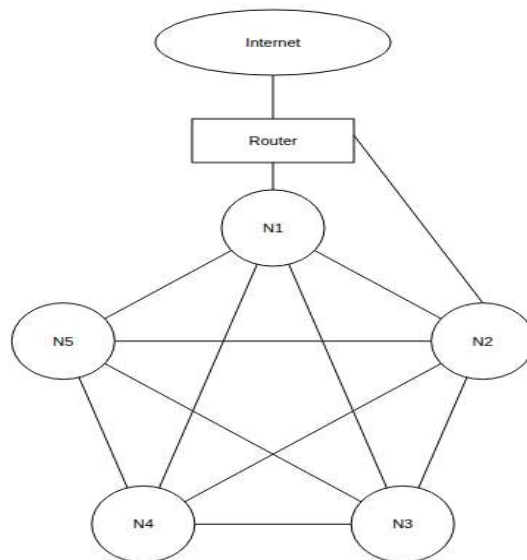
Bus Topology



A single shared backbone cable runs the length of the network, with terminators at each end and devices tapping into it via drop lines.

Real-world example: Bus topology fits an old-style local file-sharing setup in a very small office with just a handful of PCs, since it needs minimal cabling - but it is rarely used now because one break in the shared cable takes the entire network down, which is too risky for a live app or service.

Mesh Topology



N1 and N2 both connect to the router, so internet access survives if either one fails

Every node is directly connected to every other node, so there is no single central point of failure.

Real-world example: IPL live streaming infrastructure and large ISP backbone networks use mesh-like connections between core servers and routers, because if one path or link fails, traffic can automatically reroute through another connection - critical for handling millions of concurrent viewers without downtime.

Task 2: Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

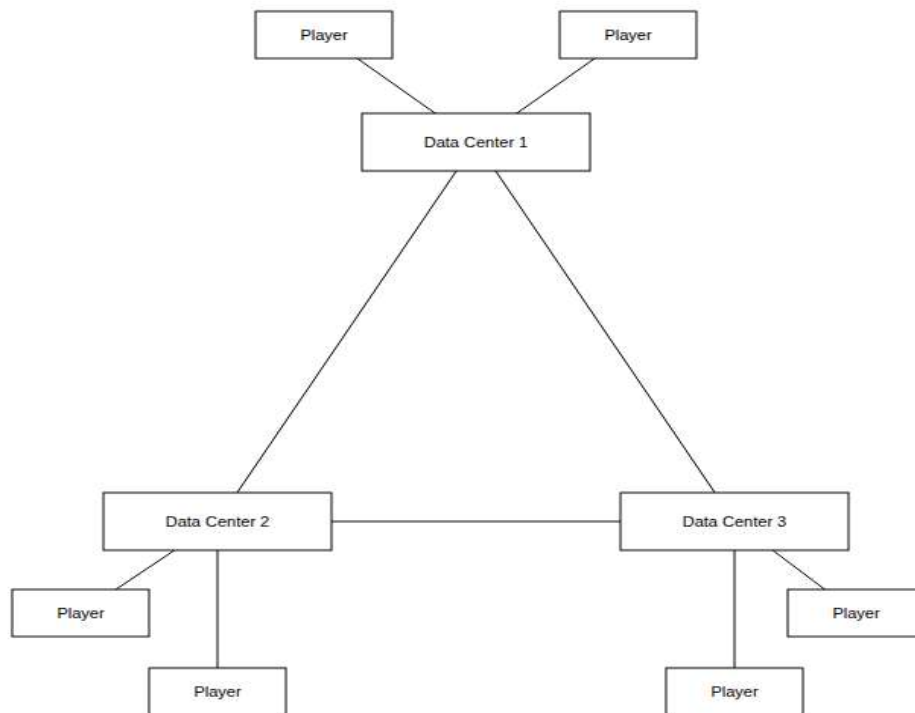
Topology	Pros	Cons	Best use case (app example)
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Star	1. Easy to add/remove devices 2. One faulty cable doesn't affect others 3. Centralized management via switch 4. Easy fault detection	1. Entire network fails if hub/switch fails 2. More cabling needed than bus 3. Cost of switch adds expense 4. Limited by switch port capacity	Office LAN for Zomato's backend team workstations - centralized, easy to manage and expand
Bus	1. Very low cabling cost 2. Simple to set up for small networks 3. Works well for a small number of devices 4. Requires less hardware overall	1. Entire network fails if backbone cable breaks 2. Difficult to troubleshoot faults 3. Performance degrades as devices increase 4. Limited cable length	A small shop's 3-4 billing counters sharing one single cable
Ring	1. Data flows in a predictable, orderly manner 2. Performs well under high traffic (no collisions) 3. Easier to identify a broken segment than bus 4. No central device dependency	1. One broken link can disrupt the entire ring (simple ring) 2. Adding/removing devices disrupts the ring 3. Harder to troubleshoot than star 4. Slower than star for large networks	Legacy token-passing setups, similar to how Instagram's internal data-center rings once used ring-based fiber backbones
Mesh	1. Highly reliable - multiple paths if one fails 2. No single point of failure 3. Excellent for high-traffic, high-priority data 4. Great security/isolation between direct links	1. Very expensive (lots of cabling) 2. Complex to install and maintain 3. Difficult to scale to many devices 4. Requires more network interfaces per device	A WhatsApp group calling app connecting every participant's device directly to every other participant

Task 3: Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

For multiplayer game servers, a star topology (with regional data centers acting as hubs) combined with a partial mesh between those data centers is the most practical choice.

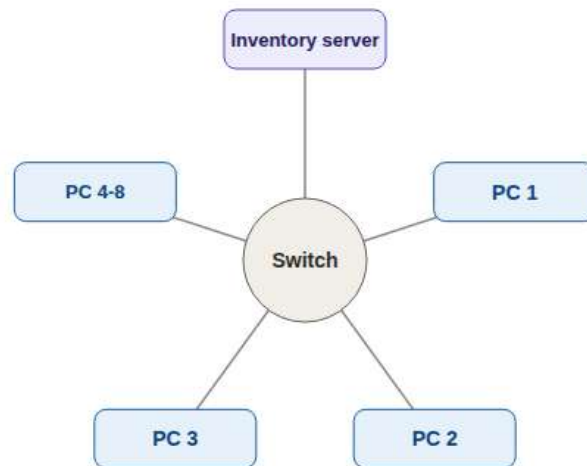
A star setup at the local server level keeps speed high, since every player's device talks directly to a nearby central game server rather than to other players individually, reducing lag. It also improves reliability, because if one player's connection drops, it does not affect anyone else's session - only the central server matters.



Each data center is a star hub for its local players; data centers are meshed for redundancy

Task 4: Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

Recommended Topology: Star



A central switch connects the inventory server and all order-processing PCs (PC 1 through PC 8) individually.

Reasoning: Star topology is ideal here because it keeps cost reasonable for just 8 machines (needing only one affordable switch), makes troubleshooting simple since a faulty cable or PC will not bring down the inventory server or the other computers, and allows easy future expansion - new computers can be added by simply plugging into a free switch port without redesigning the whole network.